Data: Current data D; historical data $D_0 = \{D_{0h} : h = 1, \ldots, H\}$; grid of points $\boldsymbol{\alpha} = \{0 = \alpha_1 < \alpha_2 < \ldots < \alpha_T = 1\}$; number of PP MCMC samples M_h; number of posterior samples M.

Result: M posterior samples under the NPP in (6).

for $h \leftarrow 1$ **to** H **do**

  for $t \leftarrow 1$ **to** T **do**

    Obtain M_h MCMC samples from the PP in (4) using historical data D_{0h}, initial prior $\pi_0^{1/H}$, and discounting parameter α_t.

    Conduct bridge sampling to obtain an estimate $\widehat{Z_h(\alpha_t)}$ of $Z_h(\alpha_t)$ using the M_h samples from the prior.

  end

  Use linear interpolation on $\{\widehat{Z_h(\alpha_1)}, \ldots, \widehat{Z_h(\alpha_T)}\}$ to obtain a piecewise linear approximation $\hat{Z}_h(\cdot)$ of $Z_h(\cdot)$.

  Draw M posterior samples under the NPP by substituting $\hat{Z}_h(\cdot)$ for $Z_h(\cdot)$ in (6).

end